Status	Finished
Started	Wednesday, 21 October 2026, 11:44 AM
Completed	Wednesday, 21 October 2026, 11:58 AM
Duration	14 mins 22 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```
s = input()
words = s.split()
result = []
for word in words:
    if word.lower() != word.lower()[::-1]:
        result.append(word)
print(" ".join(result))
```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect

Marks for this submission: 0.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```
s1 = input()
s2 = input()
n = int(input())

result = ""

for ch in s1:
    if ch in s2 and ch not in result:
        result += ch
    if len(result) == n:
        break

print(result)
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range since it does not appear in nums.
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer: (penalty regime: 0 %)

Reset answer

```
def missingNumber(nums):
    n = len(nums)
    total = n * (n + 1) // 2
    return total - sum(nums)
```

	Test	Expected	Got	
✓	`print(missingNumber([3,0,1]))`	2	2	✓
✓	`print(missingNumber([0,1]))`	2	2	✓
✓	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

You must write a solution in `O(log(m * n))` time complexity.

Example 1:

```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:

```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	`False`
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	`True`

Answer: (penalty regime: 0 %)

Reset answer

```python
def searchMatrix(matrix, target):
    m = len(matrix)
    n = len(matrix[0])
    left = 0
    right = m * n - 1

    while left <= right:
        mid = (left + right) // 2
        value = matrix[mid // n][mid % n]

        if value == target:
            return True
        elif value < target:
            left = mid + 1
        else:
            right = mid - 1

    return False
```

	Test	Expected	Got	
✓	`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	`False`	`False`	✓

	Test	Expected	Got	
✓	`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	`True`	`True`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 < nums[i], target < 10^4`
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```
def search(nums, target):
    left = 0
    right = len(nums) - 1

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	`4`	`4`	✓
✓	`print(search([-1,0,3,5,9,12],2))`	`-1`	`-1`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

```
10 6
```

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```
n = int(input())
a = list(map(int, input().split()))

peaks = []

for i in range(n):
    if (i == 0 or a[i] >= a[i - 1]) and (i == n - 1 or a[i] >= a[i + 1]):
        peaks.append(a[i])

print(*peaks)
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4 12 3 6 8	12 8	12 8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```
n = int(input())
nums = list(map(int, input().split()))
k = int(input())

seen = set()
found = False

for num in nums:
    if k - num in seen or (num * 2 == k and nums.count(num) >= 2):
        found = True
        break
    seen.add(num)

print("Yes" if found else "No")
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```
s1 = input()
s2 = input()

result = ""

for ch in s1:
    if ch not in s2:
        result += ch

print(result)
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```
nums = sorted(map(int, input().split(',')))
target = int(input())

left = 0
right = len(nums) - 1
found = False

while left <= right:
    mid = (left + right) // 2

    if nums[mid] == target:
        found = True
        break
    elif nums[mid] < target:
        left = mid + 1
    else:
        right = mid - 1

print(found)
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```
def BalancedStrings(s):
    count = 0
    balance = 0

    for ch in s:
        if ch == 'L':
            balance += 1
        else:
            balance -= 1

```

```
        if balance == 0:
            count += 1

    return count
```

	Test	Expected	Got	
✓	print(BalancedStrings('RLRRLLRLRL'))	4	4	✓
✓	print(BalancedStrings('RLLLLRRRLR'))	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Status	Finished
Started	Wednesday, 21 October 2026, 11:59 AM
Completed	Wednesday, 21 October 2026, 12:04 PM
Duration	5 mins 4 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 100
- c. 1000
- d. 10 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n ✓
- c. 1
- d. n/2

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By comparing each element sequentially
- c. By using a hash function
- d. By dividing the list length by two ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues in the left or right sublist ✓
- c. The search continues from the beginning
- d. The search stops

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There should be direct access to the middle element in any sublist
- b. Number values should only be present

  Number values should only be present

- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. The list must be sorted

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search starts over from the beginning
- b. The search backtracks to find duplicate elements
- c. The search continues until the end of the list
- d. The search stops immediately ✓

Question **7**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the left sublist

Question **8**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Interpolation search
- b. Hash search ✗
- c. Binary search
- d. Linear search

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Dividing the list in half repeatedly
- b. Checking each element sequentially ✓
- c. Skipping every second element
- d. Sorting the list before searching

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list is sorted
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It compares elements sequentially
- b. It can be applied only if the list is sorted ✓
- c. It works on unsorted lists
- d. It always starts from the beginning of the list

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Discovering
- c. Finding
- d. Mining

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By using a hash function
- b. By comparing each element in the list sequentially ✓
- c. By dividing the list into halves
- d. By sorting the list first

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It can only be applied to large lists
- d. It is slower than linear search for small lists

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✕
- b. The middle element is chosen randomly
- c. The higher middle element is chosen as the middle element
- d. The search stops